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Nutrition & Digestion

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- They are known as autotrophs. Ex: green plants ii) Heterotrophic nutrition → It is a type of nutrition in which organisms depend upon other organisms for food to survive. Ex: animals, humans - In heterotrophic also, we see : a) Saprophytic: Obtaining food from dead and decaying matter. Ex: Fungi b) Parasitic: Obtaining food from host organism without killing it. Ex: Roundworm c) Holozoic: Having solid matter and then digesting it is holozoic mode of nutrition. Ex: Animals ♦ The process by which green plants make their own food from carbon dioxide and water by using sunlight energy in the presence of chlorophyll, is called photosynthesis. ♦ Carbohydrates are utilised for providing energy to the plant. $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (Glucose) • Oxygen is released during photosynthesis. • The process of photosynthesis takes place in the green leaves of a plant. • The food is prepared by the green leaves of a plant in the form of a simple sugar called glucose. • The extra glucose is changed into another food called starch.

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- This starch is stored in the leaves of the plant. • The green plants convert sunlight energy into chemical energy by making carbohydrates. • The carbon dioxide gas enters the leaves of the plant through the stomata present on their surface. Obtaining water for Photosynthesis • The water required by the plants for photosynthesis is absorbed by the root of the plants from the soil through the process of osmosis. • The water absorbed by the roots of the plants is transported upward through the xylem vessels to the leaves where it reaches the photosynthetic cells. Site of Photosynthesis • The site of photosynthesis in a cell of the leaf are chloroplasts which contain chlorophyll. • Chloroplasts are present in the photosynthetic cells (mesophyll cells) of green plants. - These cells contain more chlorophyll than other plant cells. Stomata 1) They allow the exchange of gases (CO_2 and O_2) with the atmosphere. 2) Evaporation of water from the leaf surface occurs through the stomata. - Thus, the stomata help in the process of transpiration. 3) Allows the intake of carbon dioxide and to give out oxygen during the process of photosynthesis.

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* How do organisms obtain their nutrition? → The process of nutrition involves the following: * Ingestion → Digestion → Absorption → Assimilation → Egestion
Answers 1) All the body cells are not in direct contact with the surrounding environment, like humans in multicellular organisms. -Hence, each cell of the body do not get oxygen according to the requirement by the process of diffusion from environment. - So, diffusion is insufficient to meet the oxygen demand of multicellular organisms. 2) Breathing and respiration are the main criteria used to decide whether something is alive. - Though, living beings also show growth and movement. 3) An organism uses organic molecules as raw material. - Whereas, food is used by Heterotrophs and carbon dioxide, minerals, water is used by autotrophs. 4) Following process are essential for maintaining life: i) Nutrition ii) Respiration iii) Transportation iv) Excretion

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Ingestion → Amoeba ingests food with the help of its finger-like extensions, called pseudopodia. When a food particle approaches Amoeba, it forms pseudopodia around it and forms a food vacuole inside the Amoeba. ii) Digestion → Various enzymes from the cytoplasm enter into the food vacuole and break them down into simple soluble molecules. iii) Absorption → The simple soluble food is absorbed by cytoplasm of Amoeba from food vacuoles through the process of diffusion. iv) Assimilation → Amoeba cell obtains energy from the absorbed food through respiration. -This energy is utilised by Amoeba for its growth and repair of the body. v) Egestion → When a considerable amount of undigested food gets collected inside Amoeba, its cell membrane ruptures and throws out the undigested food. * Important Definitions ① Ingestion: The process of intake of food by an organism is called Ingestion. ② Digestion: The process in which food containing large, insoluble molecules is broken down into small, water soluble molecules.

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③Absorption: The process in which the digested food passes through the intestinal wall into blood stream is called Absorption. ④Assimilation: The process in which the absorbed food is taken in by body cells and used for energy, growth and repair is called Assimilation. ⑤Egestion (Excretion): The process in which the undigested food is removed from the body is called Egestion. ■Digestion begins in the mouth itself. 1) Mouth →Digestion begins in the mouth itself. The saliva contains an enzyme called salivary amylase that breaks down starch which is a complex molecule into sugars like maltose. 2) Buccal or Oral Cavity →It extend from mouth to pharynx and lies between two jaws, upper (fixed)

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and lower (movable). -It contains teeth also. -Teeth help in chewing and grinding the food. Starch gets converted to simple sugar/ maltose. -The movement of tongue turns food into mass of a substance called BOLUS. 3) Oesophagus →No digestion of food takes place here. -Peristaltic movement helps to pass the food down. 4) Stomach →The muscular walls of the stomach help in mixing the food thoroughly with more digestive juices. -These digestion functions are taken care by the gastric glands present in the wall of the stomach. Gastric Glands ■Gastric Glands secretes gastric juice which consists of HCl, mucus and pepsinogen. a) Function of HCl : 1) Converts inactive pepsinogen into active pepsin. 2) Kills the germs. b) function of Mucus: →Protects the inner lining of stomach. c) Function of Pepsin : →Breaks down proteins into peptides. Pepsin is a major enzyme of gastric juice which functions in acidic medium.

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